12.434

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Question

The two sides of a parallelogram are given by the vectors $\mathbf{A} = 2\hat{i} - 3\hat{j}$ and $\mathbf{B} = 3\hat{i} + 2\hat{j}$. The area of the parallelogram is:

Solution

Let:

$$\mathbf{a} = \begin{pmatrix} 2 \\ -3 \\ 0 \end{pmatrix} \tag{1}$$

$$\mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} \tag{2}$$

Solution

The area of the parallelogram can be expressed as:

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} |\mathbf{A}_{23}\mathbf{B}_{23}| \\ |\mathbf{A}_{31}\mathbf{B}_{31}| \\ |\mathbf{A}_{12}\mathbf{B}_{12}| \end{pmatrix} \tag{3}$$

Substituting values:

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ 13 \end{pmatrix} \tag{4}$$

Solution

Therefore, the area of the parallelogram is:

$$\|\mathbf{a} \times \mathbf{b}\| = 13 \tag{5}$$

It can be seen that $\mathbf{a}$ and $\mathbf{b}$ are orthogonal, and $\|\mathbf{a}\| = \|\mathbf{b}\|$, therefore they form the sides of a square.
Therefore the area can also be calculated as:

$$\|\mathbf{a}\|^2 = (\sqrt{13})^2 = 13 \tag{6}$$

Code

Codes permalink

Figure

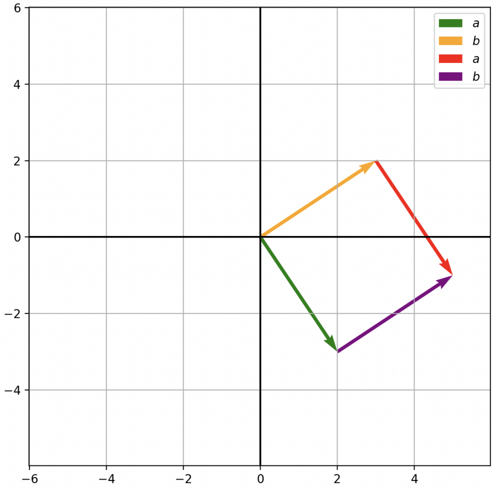

Figure: Plot